



Troubleshooting-CUCM

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Menu

1 CUCM performance

2.CUCM Call Troubleshooting

High level the troubleshooting steps

1. Validate VM Deployment
2. Troubleshoot performance at Guest-OS Level
(Application itself)
3. Troubleshoot performance at Host ESXi level

Validate VM Deployment

Validate VM deployment

The deployment must meet the TRC (Tested Reference Configuration) specifications.

Here are key documents that can be referenced when validating VM deployment:

[Cisco Collaboration on Virtual Servers](https://www.cisco.com/c/en/us/td/docs/voice_ip_comm/cucm/virtual/CHCS_BK_C7C7ED05_00_cisco-collaboration-on-virtual-servers/CHCS_BK_C7C7ED05_00_cisco-collaboration-on-virtual-servers_chapter_0100.html)

https://www.cisco.com/c/en/us/td/docs/voice_ip_comm/cucm/virtual/CHCS_BK_C7C7ED05_00_cisco-collaboration-on-virtual-servers/CHCS_BK_C7C7ED05_00_cisco-collaboration-on-virtual-servers_chapter_0100.html

[Cisco Collaboration Virtualization](https://www.cisco.com/c/dam/en/us/td/docs/voice_ip_comm/uc_system/virtualization/cisco-collaboration-virtualization.htm)

https://www.cisco.com/c/dam/en/us/td/docs/voice_ip_comm/uc_system/virtualization/cisco-collaboration-virtualization.htm

[Collaboration Virtualization Hardware](https://www.cisco.com/c/dam/en/us/td/docs/voice_ip_comm/uc_system/virtualization/cisco-collaboration-infrastructure.html)

https://www.cisco.com/c/dam/en/us/td/docs/voice_ip_comm/uc_system/virtualization/cisco-collaboration-infrastructure.html

Validate VM deployment

The primary method for validating VM deployment specifications via the VOS-based application is to run cli command: **show hardware**.

VM Deployment Validation via ESXi Vsphere Client

To verify that the VM specs adhere to documentation, you can navigate Vpsphere Web client and manually check CPU, Memory, and Disk requirements

```
admin:show hardware
```

```
HW Platform      : VMware Virtual Machine
Processors       : 1
Type             : Intel(R) Xeon(R) Gold 6132 CPU @ 2.60GHz
CPU Speed        : 2600
Memory           : 6144 MBytes
Object ID        : 1.3.6.1.4.1.9.1.1348
OS Version       : UCOS 7.0.0.0-4.i386
Serial Number    : VMware-42 13 79 13 d5 e3 69 36-37 b6 d7 7f bf bf 4d cb
```

```
RAID Version     :
No RAID controller information is available
```

```
BIOS Information :
PhoenixTechnologiesLTD 6.00 04/05/2016
```

```
RAID Details     :
No RAID information is available
```

```
-----
Physical device information
-----
```

```
Number of Disks   : 1
Hard Disk #1
Size (in GB)      : 80
```

```
Partition Details :
```

```
Disk /dev/sda: 10443 cylinders, 255 heads, 63 sectors/track
Units: sectors of 512 bytes, counting from 0
```

Device	Boot	Start	End	#sectors	Id	System
/dev/sda1	*	2048	29028351	29026304	83	Linux
/dev/sda2		29028352	58054655	29026304	83	Linux
/dev/sda3		58054656	58578943	524288	83	Linux
/dev/sda4		58578944	167772159	109193216	5	Extended
/dev/sda5		58580992	66772991	8192000	82	Linux swap / Solaris
/dev/sda6		66775040	167772159	100997120	83	Linux

Troubleshooting Performance at Guest-OS Level (VOS Platform)

Troubleshooting Performance at Guest-OS Level (VOS Platform)

- Troubleshooting steps:
 - Identify the process that's causing the high cpu condition.
 - Try to find the reason why the specific process is consuming the cpu (could be due to mis-configuration or a bug).

Troubleshooting Performance at Guest-OS Level (VOS Platform)

Collect the deployment detail and Identify the offending process(s):

- Get Deployment/Environment information:

- show version active
 - show hardware
 - show status
 - show network cluster

- Get outputs for Performance-Related CLI Commands:

- show process load [cpu | memory]
 - show process using-most [cpu | memory]
 - show perf query class "Cisco Tomcat Connector"
 - show perf query class "Cisco Tomcat Web Application"
 - show perf query class Memory
 - show perf query class Process

Troubleshooting Performance at Guest-OS Level (VOS Platform)

The primary mechanism for troubleshooting Guest-OS level CPU issues will be output from CLI command: **show process load** (equivalent to Linux "top")

```
admin: show process load cpu
top - 12:20:26 up 105 days, 4:06, 3 users, load average: 0.85, 0.82, 0.77
Tasks: 219 total, 1 running, 217 sleeping, 0 stopped, 1 zombie
Cpu(s): 6.2% us, 4.0% sy, 0.0%ni, 89.7% id, 0.1% wa, 0.0% hi, 0.0% si, 0.0% st
Mem: 3909524k total, 3764516k. used, 145008k free, 254000k buffers
Swap: 4095996k total, 612648k used, 3483348k free, 1497952k cached

PID      USER      PR    NI   VIRT   RES     SHR S  %CPU  %MEM    TIME+  COMMAND
59953    tomcat    20     0   2510m  535m    13m S   2.0   14.0   1364:34 tomcat
62578     root     20     0    2624    660     596 S   2.0    0.0   19:12.91 hostagt
125960   admin     30    10    719m   49m     19m S   2.0    1.3    0:03.21 java
1         root     20     0   19572   1692    1200 S   0.0    0.0   11:48.38 init
```

Troubleshooting Performance at Guest-OS Level (VOS Platform)

To evaluate CPU performance and the offending process, focus on the the "load average", the %CPU values, the Cpu(s) statistics row of values

```
admin: show process load cpu
top - 12:20:26 up 105 days, 4:06, 3 users, load average: 0.85, 0.82, 0.77
Tasks: 219 total, 1 running, 217 sleeping, 0 stopped, 1 zombie
Cpu(s): 6.2% us, 4.0% sy, 0.0%ni, 89.7% id, 0.1% wa, 0.0% hi, 0.0% si, 0.0% st
Mem: 3909524k total, 3764516k. used, 145008k free, 254000k buffers
Swap: 4095996k total, 612648k used, 3483348k free, 1497952k cached
```

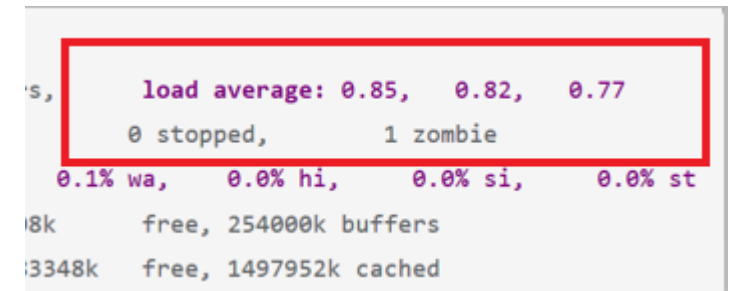
PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
59953	tomcat	20	0	2510m	535m	13m	S	2.0	14.0	1364:34	tomcat
62578	root	20	0	2624	660	596	S	2.0	0.0	19:12.91	hostagt
125960	admin	30	10	719m	49m	19m	S	2.0	1.3	0:03.21	java
1	root	20	0	19572	1692	1200	S	0.0	0.0	11:48.38	init

Troubleshooting Performance at Guest-OS Level (VOS Platform)

- **Load Average Interpretation**

The example represents a single-core (1 vCPU) system, meaning that only 1 process can run at any given time. The load average numbers mean

- over the last 1 minute, the CPU idles for 15% of the time (utilized for 85%)
- over the last 5 minutes, the CPU idled for 18% of the time (utilized for 82%)
- over the last 15 minutes, the CPU idled for 23% of the time (utilized for 77%)



What if the numbers for a single-core system instead read: **load average: 1.25, 0.92, 3.75**

- over the last 1 minute, the CPU was overloaded by 25% on average meaning that 0.25 processes were waiting for CPU
- over the last 5 minutes, the CPU idled for 8% of time (utilized for 92%)
- over the last 15 minutes, the CPU was overloaded by 275% on average meaning that 2.75 processes were waiting for CPU

Troubleshooting Performance at Guest-OS Level (VOS Platform)

- **REMEMBER:** when interpreting load average, the number of vCPU must always be considered
- **General interpretations**
 - If the averages are 0.0, then your system is idle.
 - If the 1 minute average is higher than the 5 or 15 minute averages, then load is increasing.
 - If the 1 minute average is lower than the 5 or 15 minute averages, then load is decreasing.
 - If they are higher than your CPU count, then you might have a performance problem

Troubleshooting Performance at Guest-OS Level (VOS Platform)

CPU% Interpretation

- To calculate the overall system CPU utilization from the output = (sum of all %CPU) / (# of vCPU)

PID	USER	PR	NI	VIRT	RES	SHR S	%CPU	%MEM	TIME+	COMMAND
59953	tomcat	20	0	2510m	535m	13m S	2.0	14.0	1364:34	tomcat
62578	root	20	0	2624	660	596 S	2.0	0.0	19:12.91	hostagt
125960	admin	30	10	719m	49m	19m S	2.0	1.3	0:03.21	java
1	root	20	0	19572	1692	1200 S	0.0	0.0	11:48.38	init

In the above example, there is 6% active CPU utilization because $(2.0 + 2.0 + 2.0) / 1 = 6$.

NOTE: this is a single vCPU IMP server

Troubleshooting Performance at Guest-OS Level (VOS Platform)

Cpu(s) Statistics Row

This row of data shows the percentage of CPU time spent on various tasks.

Here is break down of what the percentages mean across various task types, along with troubleshooting t

- **US (user)** - CPU time spent executing user-space processes.

These are higher-level processes, like your application, or the database server running on your machine.

- **SY (system)** - CPU time spent executing kernel-space processes.

The kernel is responsible for low-level tasks, like interacting with the hardware, memory allocation, communicating between OS

- **NI (nice)** - the priority level a user space process can be tweaked by adjusting its *niceness*. **ID (idle)** - The “idle” CPU state

- **WA (iowait)** - “iowait” is a sub category of the “idle” state.

- It marks **time spent waiting for input or output operations**, like reading or writing to disk.
- When the processor waits for a file to be opened, for example, the time spend will be marked as “iowait”.
- Elevated CPU time in the “iowait” category can reveal problems outside of the processor.
- For example, when an in-memory database needs to flush a lot of data to disk, or when memory is swapped to disk.

Troubleshooting Performance at Guest-OS Level (VOS Platform)

- Other outputs

- utils core active list
 - utils diagnose test
 - utils service list

- Collect the following minimum log set

- RTMT: Event Viewer Application, **CLI:** file get activelog syslog/CiscoSyslog*, **Root:** /var/log/active/syslog/CiscoSyslog*
 - RTMT: Event Viewer System, **CLI:** file get activelog syslog/messages* , **Root:** /var/log/active/syslog/messages*
 - RTMT: Cisco RIS Data Collector Perfmon Log
 - RTMT: Service identified as cause of the high CPU

Troubleshoot performance at Host ESXi level

Troubleshoot performance at Host ESXi level

ESXi Embedded Vsphere Client

To view ESXi-host performance, view hardware events, etc... go to **Host > Monitor > Overview**. From here you can select the appropriate tab and view the desired information.

To view specific VM-performance info, click the "Monitor" link on VM Overview page.

Troubleshoot performance at Host ESXi level

Vcenter Vsphere Client:

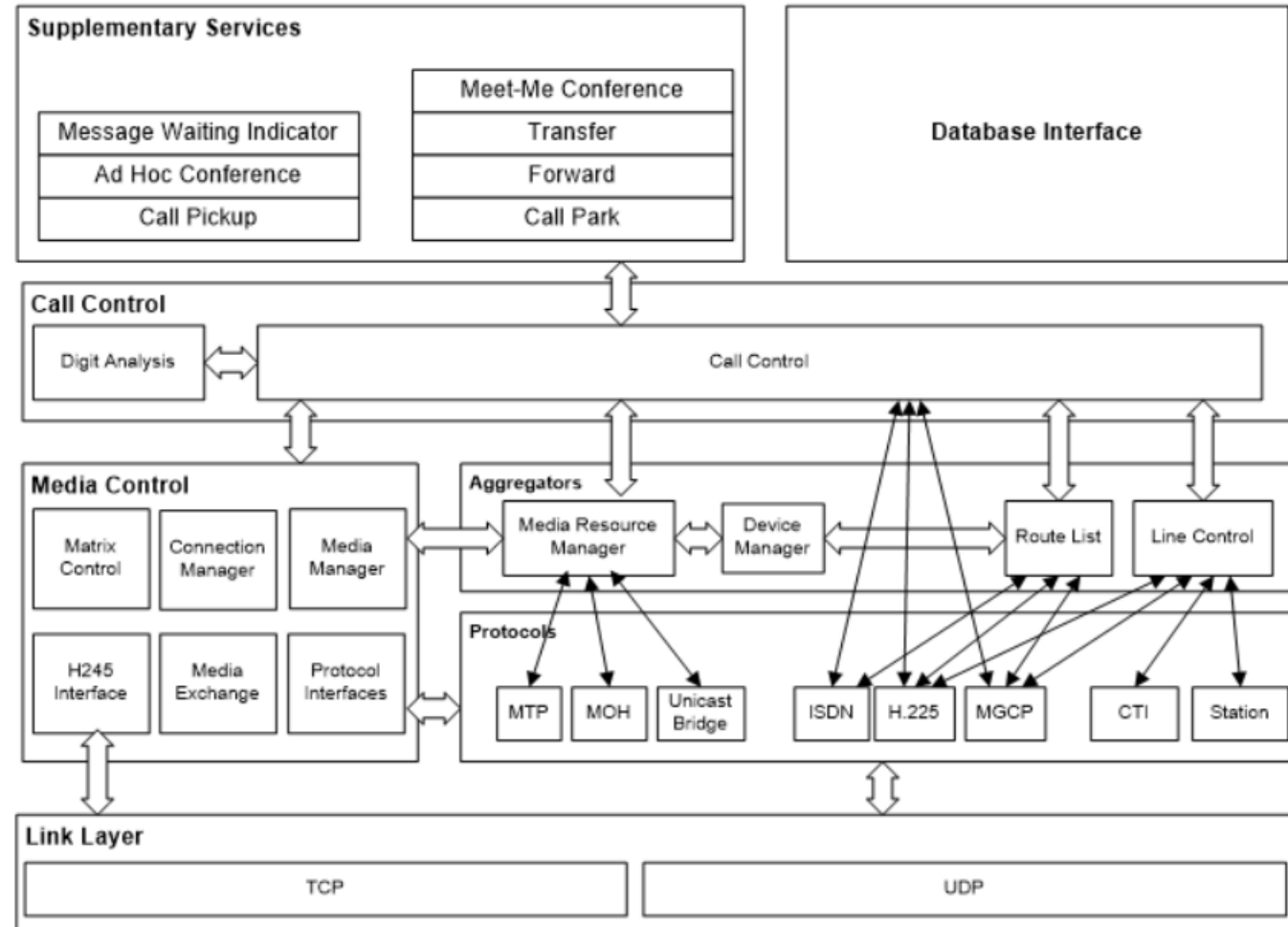
Select Host_or_VM > Monitor > Performance >
Overview

CUCM Call Troubleshooting

1. IP phone to SIP trunk
2. Media resource

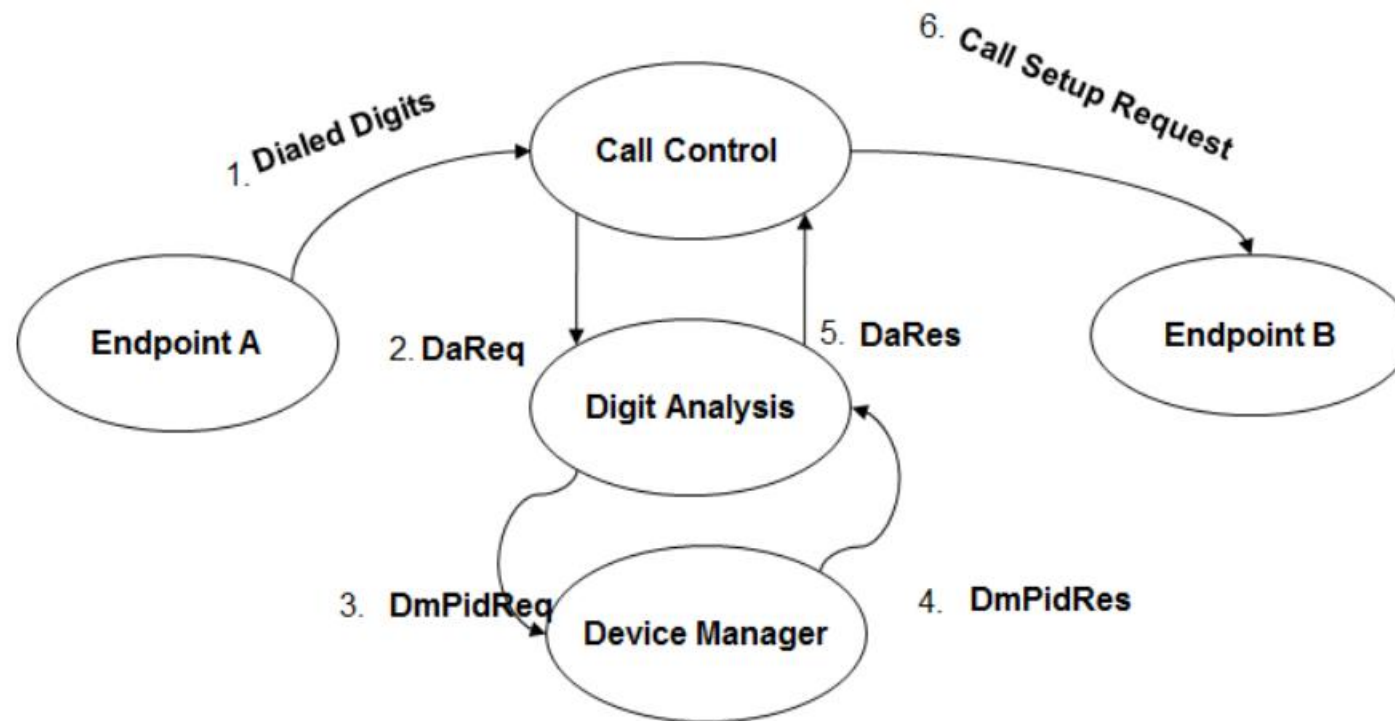
CUCM Block Diagram

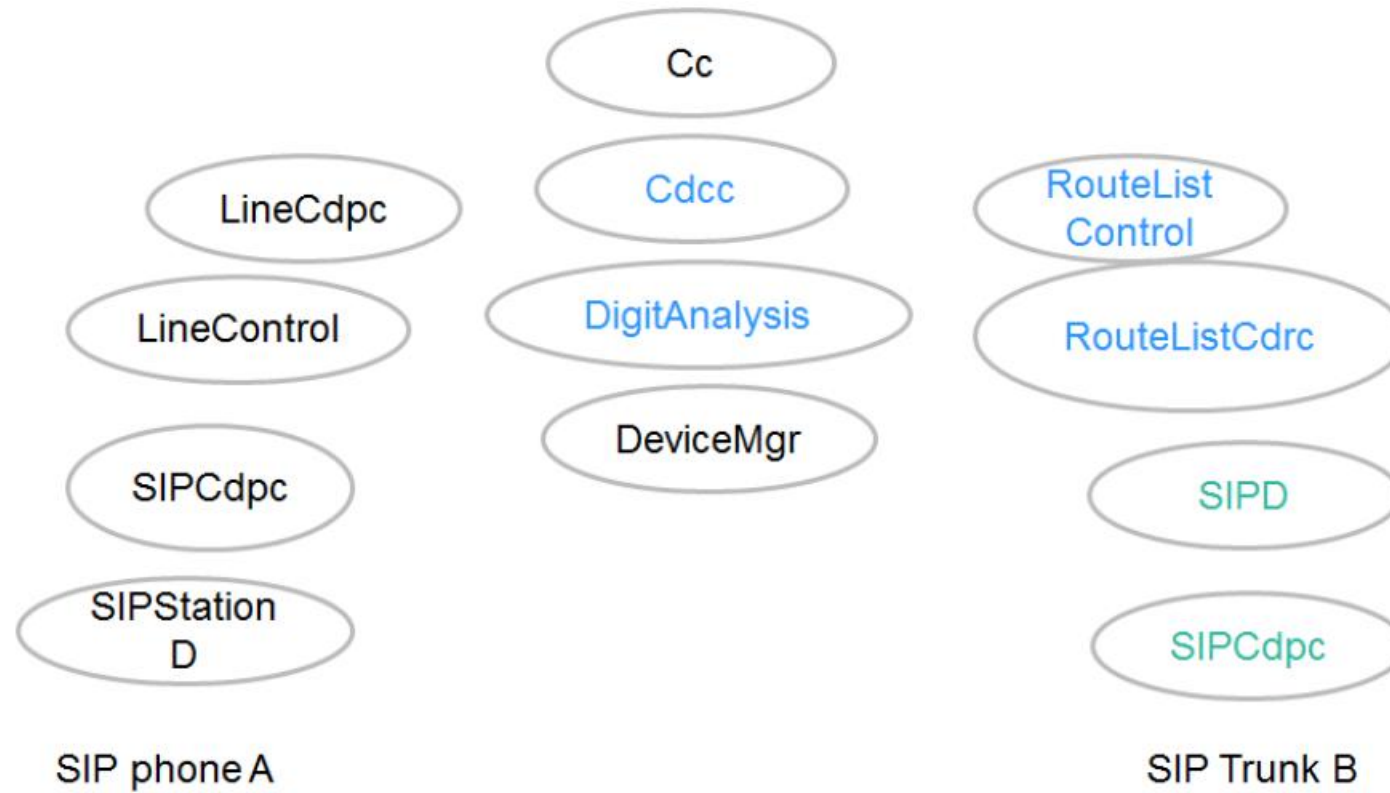
Cisco CallManager Block Diagram



- CC (one per node)
- Cdcc (one per call)
- DigitAnalysis (matches valid dialed patterns)
- DeviceManager (maps valid pattern to the pid of its controlling process)
- LineControl (controls one line which may appear on multiple devices)
- RouteListControl (manages on routelist which is associated with a dialed routepattern. Each routelist may contain several route groups. Each route group is associated with a list of destination devices such as unity ports, gateways or trunks)

- StationInit (interfaces with all skinny devices)
- SIPStationInit (interfaces with all sip phones)
- StationD (controls one skinny phone/unity port – TCP handle)
- SipStationD (controls one sip phone)
- SipD (controls one SIP Trunk)
- H225D (controls one H323 gateway)
- MGCPpn9D (controls one MGCP PRI gateway)
- MGCPTrunkD (controls one T1-CAS MGCP gateway)





15120734.000 |16:25:17.114
|SdISig |DaReq |wait |Da(4,100,2
11,1) |Cdcc(4,100,219,20) |4,100,14,143225.29^10
.143.10.159^CSF199005 [[R:N-H:0,N:0,L:0,V:0,Z:0,D:0] CI=72520520
Fqdn=ti=1nd=199005pi=0si1

15120734.013 |16:25:17.114 |ApplInfo |Digit analysis: analysis results
15120734.014 |16:25:17.114
|ApplInfo ||PretransformCallingPartyNumber=199005
|CallingPartyNumber=199005
|DialingPartition=Rendezvous_PT
|DialingPattern=80XXXX
|FullyQualifiedCalledPartyNumber=800401
(省略)
|ResourcePriorityNamespace=
|PatternRouteClass=RouteClassDefault



15120735.000 |16:25:17.114

|SdlSig |DmPidReq |initialized |DeviceManager(4,100,205,1) |Da(4,100,211,1)
|4,100,14,143225.29^10.143.10.159^CSF199005 |[T:N-H:0,N:0,L:0,V:0,Z:0,D:0] Cepn=5d41ff72-833f-d4bc-89ae-462a223be396 Id=3823371096

15120736.000 |16:25:17.114

|SdlSig |DmPidRes |wait |Da(4,100,211,1) |DeviceManager(4,100,205,1)
|4,100,14,143225.29^10.143.10.159^CSF199005 |[R:N-H:0,N:0,L:0,V:0,Z:0,D:0] Cepn=5d41ff72-833f-d4bc-89ae-462a223be396 Id=3823371096 ccmType=3 DeviceName=Rendezvous_RL Pid=4,100,89,6,

15120737.000 |16:25:17.114

|SdlSig |DaRes |setup_da |Cdcc(4,100,219,20) |Da(4,100,211,1) |4,100,14,1
43225.29^10.143.10.159^CSF199005 |[R:N-H:0,N:0,L:0,V:0,Z:0,D:0] Cl=72520520 Dial NoPotentialMatchesExist OnNet

15120747.000 |16:25:17.114

|SdlSig |CcSetupReq |idle |RouteListControl(4,100,89,6) |Cdcc(4,100,219,20) |4,100,1
4,143225.29^10.143.10.159^CSF199005 |[R:N-H:0,N:2,L:0,V:0,Z:0,D:0] Cl=72520521 Cl.branch=0 sBPL.plid=65 sBPL.l=1 sBPL.pl=5
sBPL.msdl=0

15120747.001 |16:25:17.114 |AppInfo |RouteListControl::idle_CcSetupReq - RouteList(Rendezvous_RL), numberSetup=1
numberMember=2 vmEnabled=0

```
15120747.002 |16:25:17.115 |Created | |RouteListCdrc(4,100,90,18)
|RouteListControl(4,100,89,6) | |NumOfCurrentInstances: 2
15120748.000 |16:25:17.115 |SdlSig |Start |start
|RouteListCdrc(4,100,90,18) |RouteListCdrc(4,100,90,18)
|4,100,14,143225.29^10.143.10.159^CSF199005 |*TraceFlagOverrode
```

15120748.001 |16:25:17.115 |ApplInfo |RouteListCdr::StartTransition non EMCC call
15120748.002 |16:25:17.115 |ApplInfo |RouteListCdr - RouteList Info, by RouteGroups
15120748.003 |16:25:17.115 |ApplInfo |RouteList - RouteListName="Rendezvous_RL"
CallableEndPointName="5d41ff72-833f-d4bc-89ae-462a223be396" routeListEnabled="1"

15120751.000 |16:25:17.115 |SdISig |CcSetupReq |null0
|RouteListCdrc(4,100,90,18) |RouteListControl(4,100,89,6)
|4,100,14,143225.29^10.143.10.159^CSF199005 |[R:N-H:0,N:0,L:0,V:0,Z:0,D:0] Cl=72520521 Cl.branch=0
sBPL.plid=65 sBPL.l=1 sBPL.pl=5

15120754.000 |16:25:17.115 |SdISig |CcSetupReq |restart0
|SIPD(4,100,81,14) |RouteListCdrc(4,100,90,18)
|4,100,14,143225.29^10.143.10.159^CSF199005 |[R:N-H:0,N:0,L:0,V:0,Z:0,D:0] Cl=72520521 Cl.branch=0
sBPL.plid=65 sBPL.l=1 sBPL.pl=5

15120754.006 | 16:25:17.115
| *Created* | | | *SIPCdpc(4,100,82,38)* | *SIPD(4,100,81,1*
4) | | | NumOfCurrentInstances: 4
15120755.000 | 16:25:17.115
| SdlSig | *Start* | start | *SIPCdpc(4,100,82,38)* | *SIPCdpc(4,100,*
82,38) | 4,100,14,143225.29^10.143.10.159^CSF199005 | *TraceFlagOverrode

15120756.000 |16:25:17.115 |SdlSig |CcSetupReq |null0
|SIPCdpc(4,100,82,38) |SIPD(4,100,81,14)
|4,100,14,143225.29^10.143.10.159^CSF199005 |[R:N-H:0,N:1,L:0,V:0,Z:0,D:0] CI=72520521
CI.branch=0 sBPL.plid=65 sBPL.l=1

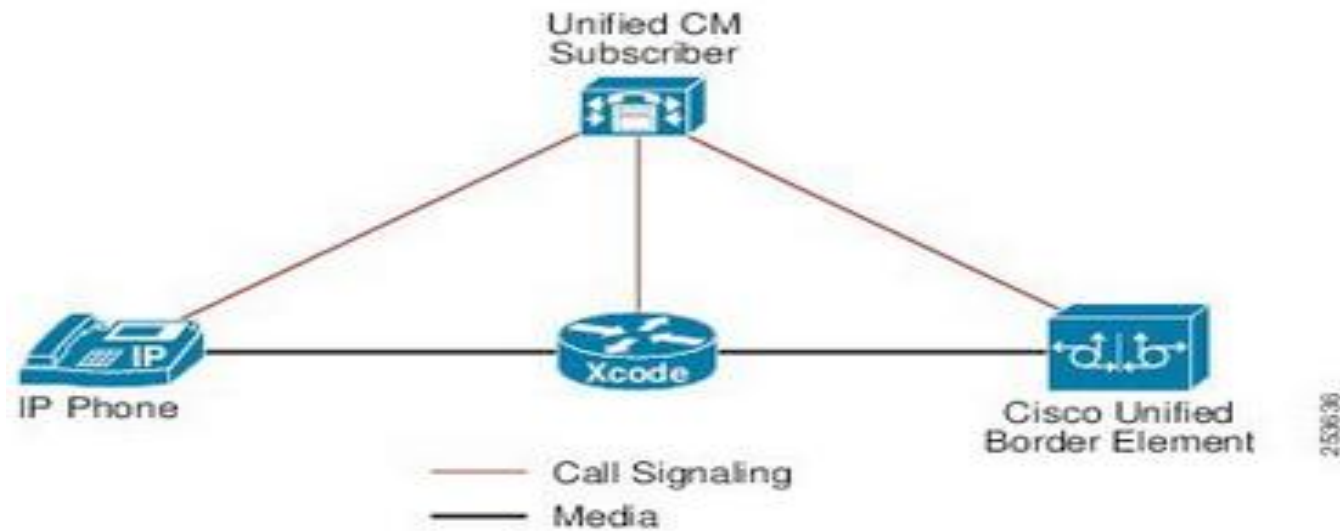
15120767.000 |16:25:17.117 |SdlSig |SIPSetupReq |wait
|SIPHandler(4,100,79,1) |SIPCdpc(4,100,82,38) |4,100,14,1.53^*^* |[T:N-
H:0,N:0,L:0,V:0,Z:0,D:0] SignalInfo= SIPInviteNoSdp Ccbld=662550 --TransType=1 --TransSecurity=0

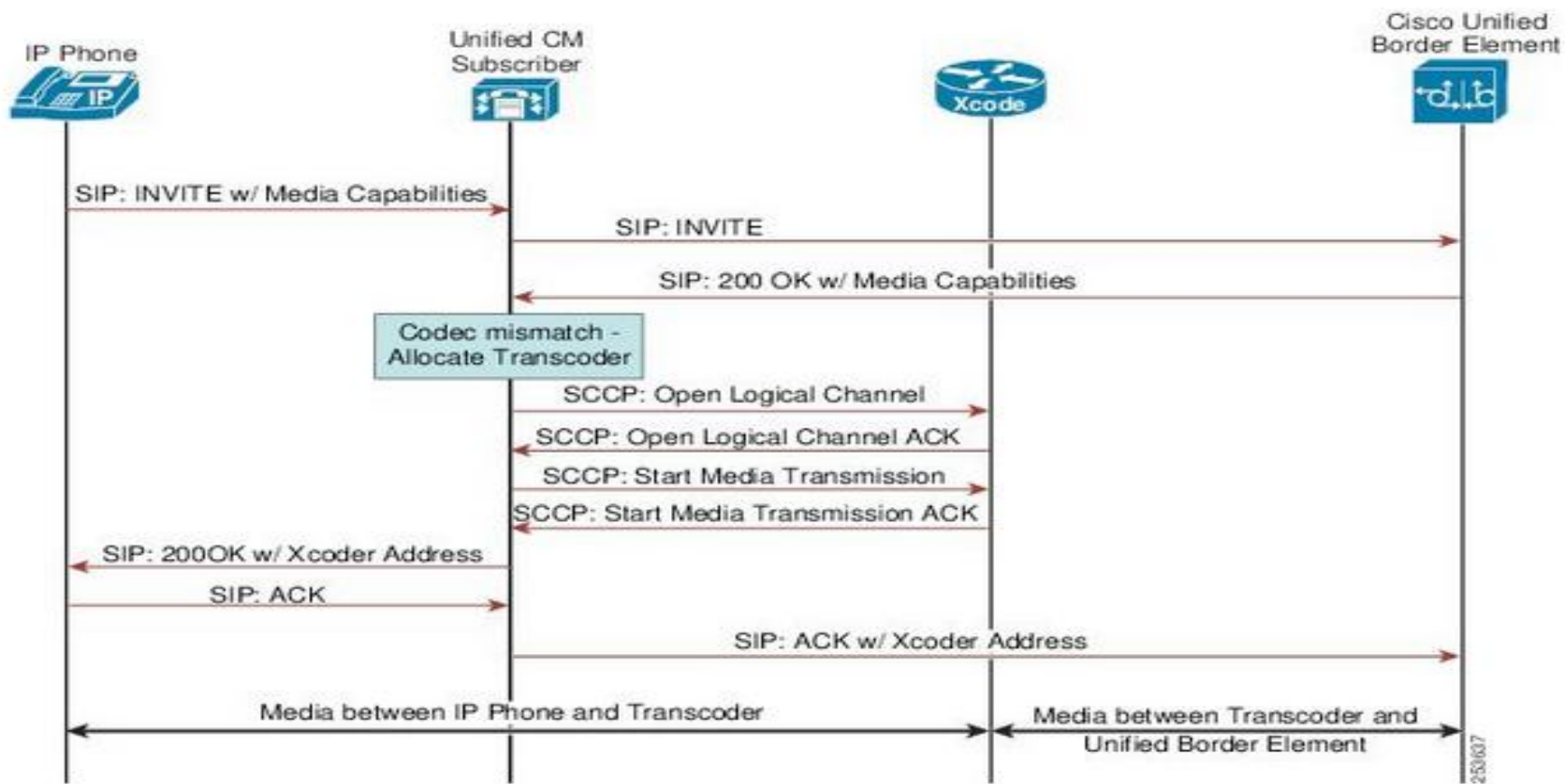
```
15120779.000 |16:25:17.117 |SdlSig |SIPSPISignal |wait  
|SIPTcp(4,100,71,1) |SIPHandler(4,100,79,1) |4,100,14,1.53^*^*  
|*TraceFlagOverrode  
15120779.001 |16:25:17.117 |AppInfo |SIPTcp - wait_SdlSPISignal: Outgoing SIP TCP message to  
172.23.3.39 on port 5060 index 10  
[1383621,NET]  
INVITE sip:800401@172.23.3.39:5060 SIP/2.0  
Via: SIP/2.0/TCP 172.23.3.31:5060;branch=z9hG4bK65cee23623fe9  
From: <sip:199005@172.23.3.31>;tag=662550~26379d3d-2d51-4405-be0b-059f34ede7fe-72520521  
To: <sip:800401@172.23.3.39>  
Date: Fri, 16 Oct 2015 07:25:17 GMT  
Call-ID: b4ef200-6201a65d-629ca-1f0317ac@172.23.3.31
```

Media resource

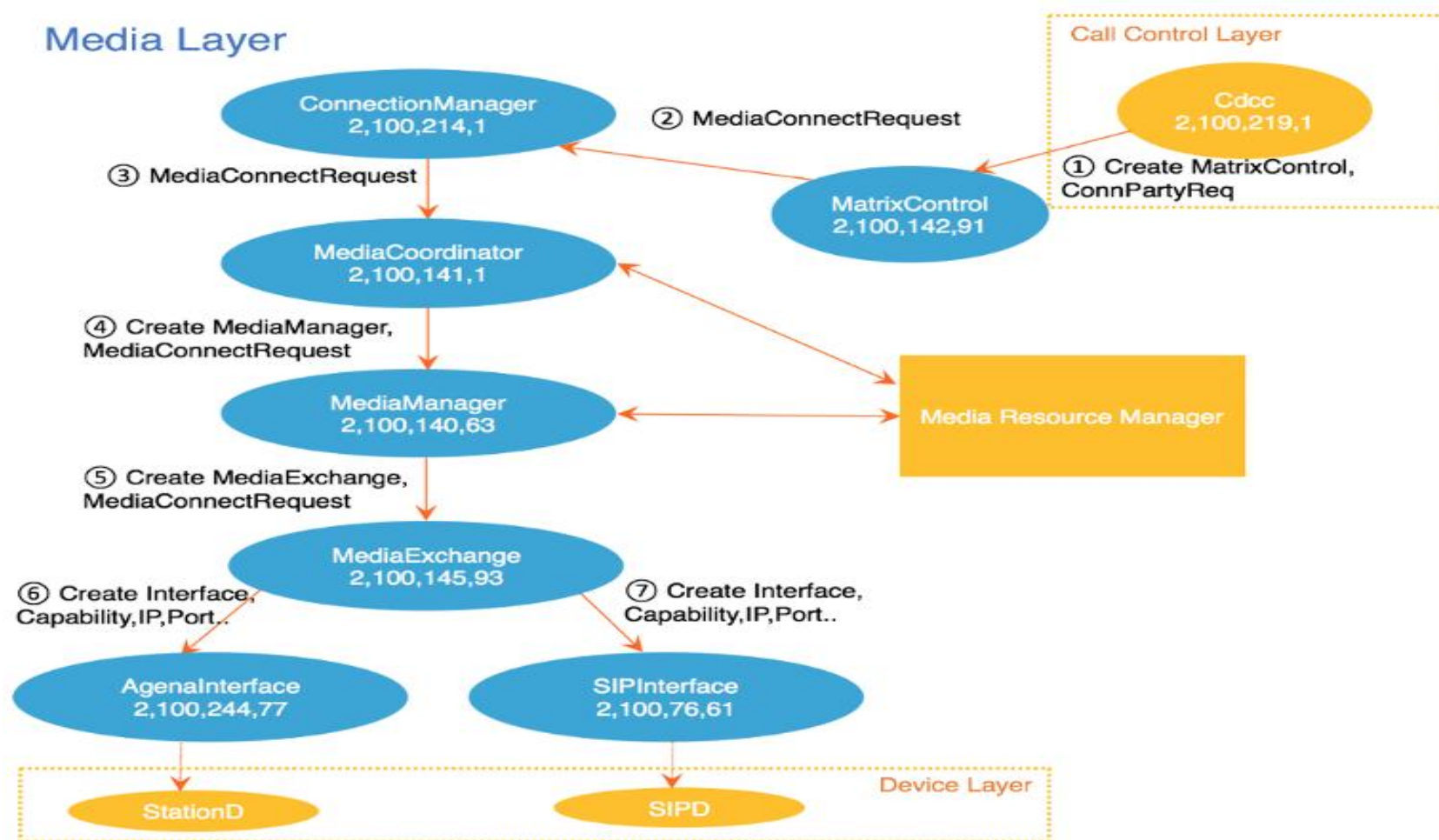
- Media Layer Role
- 1. Negotiation of multimedia capabilities between devices
- 2. Request Media Resource (XCoder, RSVP Agent, MTP (DTMF Relays, Trusted Relay Points, IPv4 / v6 Translation, Re-Packetization)) from Media Resource Manager (MRM)
- 3. Establishing a multimedia streaming channel between devices

- Call flow using Media Resource





- Media Layer operation overview



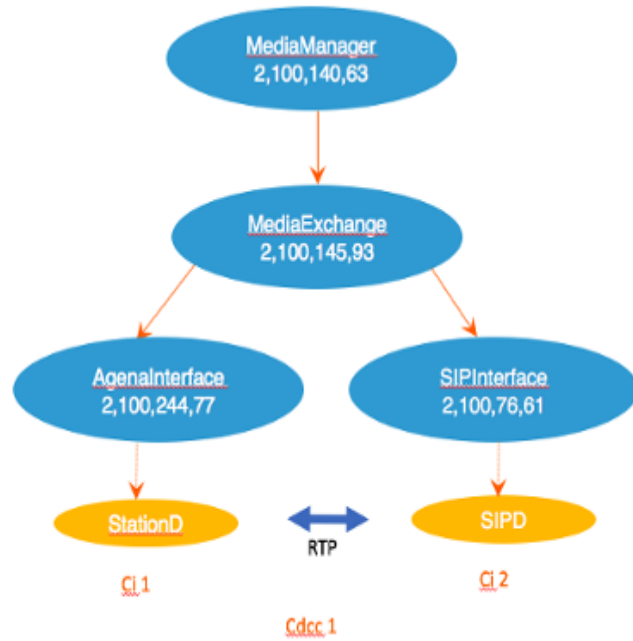
- ① Cdcc creates MatrixControl and sends
- ConnPartyReq ② ③ ④ Sends MediaConnectRequest
- ⑤ MediaManager creates MediaExchange for each RTP Hop
- (*) One MediaManager is generated per Connection. Request Media Resource from Media Resource Manager (MRM) as needed.
- ⑥ ⑦ MediaExchange creates an Interface according to the Device protocol and exchanges Media Capability, IP, Port, etc.
- (*) SIPInterface (SIP Line & Trunk) H245Interface (H323 Device, GW, Trunk),
AgenalInterface (SCCP Phone, MOH, Conference Bridge), MTP AgenalInterface (MTP),
MGCPInterface (MGCP GW)

- Media Resource
- Communicate with CUCM using SCCP
- **Music on Hold (MOH)** : Cisco IP Voice Media Streaming Application service (IPVMS)
- Annunciator (ANN) : IPVMS
- Conference Bridge (CFB): IPVMS, IOS Router, MCU, etc.

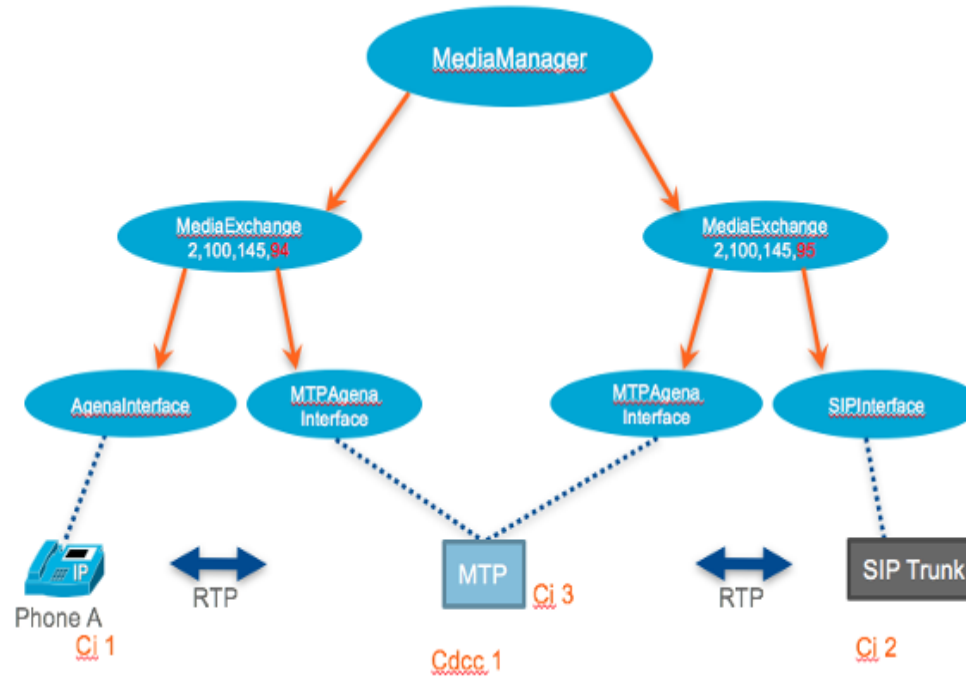
- **Media Termination Point (MTP)** : IPVMS, IOS Router
- Re-Packetization
- Used for H323 Outbound FastStart, SIP EarlyOffer
- DTMF Translation/Detection
- RSVP Agent
- TRP (VRF/Firewall Traversal, Trusted DSCP)
- IPv4/v6 Translation
- **Transcoder**

What is RTP Hop?

1 Hop (Media Resource unnecessary calls such as between IP phones Trunk calls, etc.)

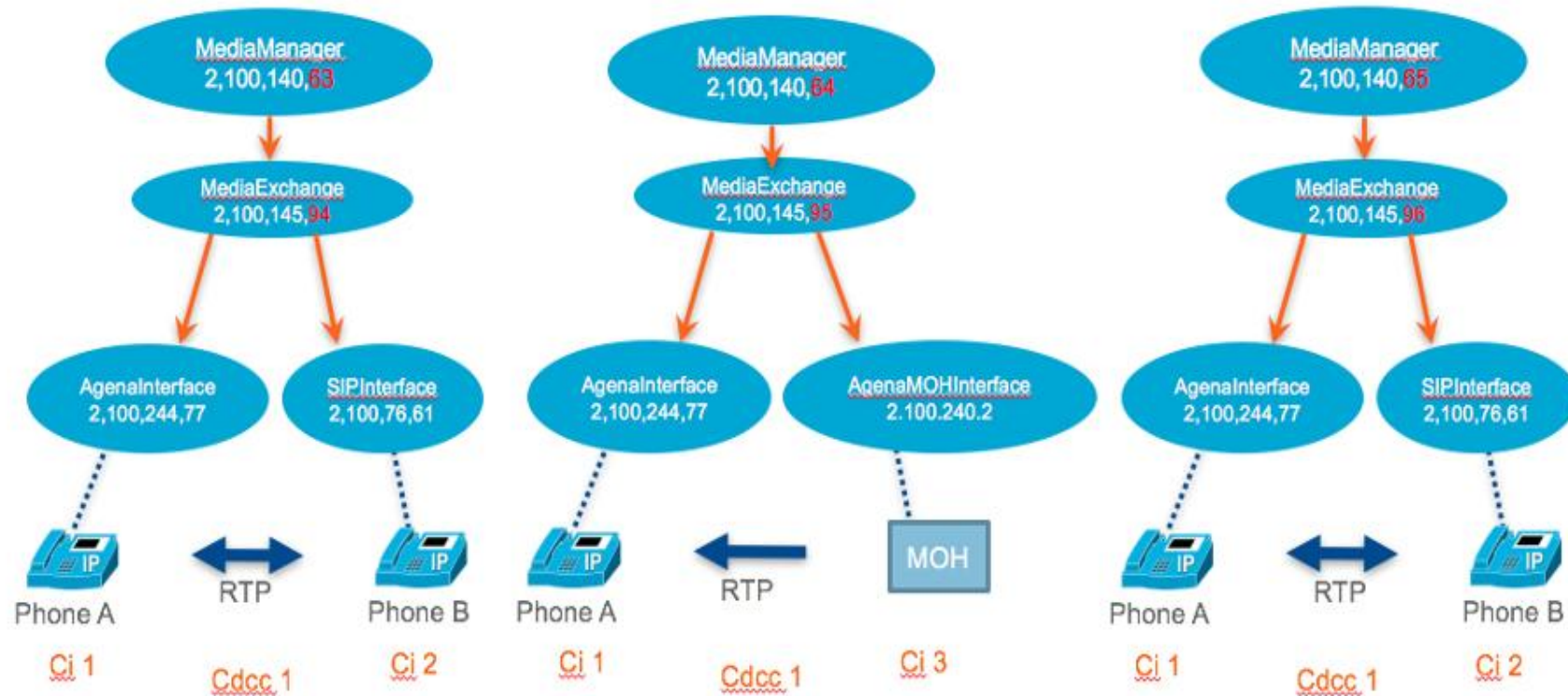


2 Hop (Calls that sandwich MTP between IP Phone and SIP



- Two MediaExchanges are generated because there are two RTPs in a call that sends with an MTP.
- [Digression] In the case of MTP, the Call Control Layer also assigns Ci for Phone A and SIP Trunk to one Cdcc.

Reservation-Reservation Release



MOH

- The settings on the hold side determine the audio file to be played, and the settings on the hold side determine the resource or server to play the file.

MTP related settings

- Early Offer Support for voice and video calls

Session Refresh Method*

Early Offer support for voice and video calls*

☐ Enable ANAT

☐ Deliver Conference Bridge Identifier

☐ ...

Invite

Disabled (Default value)

Disabled (Default value)

Best Effort (no MTP inserted)

Mandatory (insert MTP if needed)

- it was a checkbox named Early Offer Support for voice and video calls (insert MTP if needed).
- 1. if you check this, you will always get an Early Offer regardless of whether you get SDP with Initial INVITE from the calling side. He allocates MTP if he does not receive SDP
- 2. Wasted resources, MTP (and without Codec Pass Through) and video call failed, SRTP failed

- If you select Best Effort (no MTP inserted):
- When the caller receives SDP information with Initial INVITE, INVITE with SDP (Early Offer) is sent to SIP Trunk without MTP.
- If Initial INVITE does not receive SDP information from the caller, send INVITE without SDP (Delayed Offer) to SIP Trunk

- Media Termination Point Required checkbox (SIP Trunk setting)
- If checked, MTP will be allocated unconditionally
- When unchecked, whether or not MTP is used is dynamically determined based on conditions such as DTMF and Early Offer.

Transcoder

- Transcoder also has MTP functionality
- If both Transcoder and MTP are required (Codec Mismatch and DTMF Mismatch), Transcoder will be allocated.
- Transcoders may be allocated even if MTP is required, depending on Media Resource usage.
- If the Transcoder has a higher priority than the MTP in MRGL, the Transcoder will be allocated even if only the MTP is needed.

Trace Keyword

- MediaConnectRequest

// Example of a call from a SIP Phone to an Office Link via a SIP Trunk

```
00029255.000 |19:38:55.379 |SdlSig |MediaConnectRequest |waitConnectRequest |MediaManager(1,100,140,6) |MediaCoordinator(1,100,141,1)
|1,100,14,332,4^192.168.98.143^* |[R:N-H:0,N:0,L:0,V:0,Z:0,D:0]Party1: MR=0 CI=22104488 audioCapCount=7 region=DefaultxferMode=16 mrid=0 audioId=0 MMCap=0x1 sipConfig:
BFCPAllowed=T IXAllowed=T activeCap=0 cryptoCapCount=0 flushIns=0 dtm.mode=0 dtm.CI=0 dtm.MTPForDTMF=F IFPid=(0,0,0,0) dtMedia=F honorCodec=F EOType=0 DTMF Caps(1,3,101,0,F)
confID=0 connType=3 connStatus=0 mtpPre=F teleEve=0 IFCreated=F IFHandling=0 FS=0 mcNodeId=0LatentCaps=null dtm.mode=0 dtm.CI=0 dtm.MTPForDTMF=F Party2: MR=2 CI=22104489
audioCapCount=1 region=Default xferMode=16 mrid=22104490 audioId=16777223 MMCap=0x1 sipConfig: BFCPAllowed=F IXAllowed=F activeCap=0 cryptoCapCount=0 flushIns=0 dtm.mode=0
dtm.CI=0 dtm.MTPForDTMF=F IFPid=(0,0,0,0) dtMedia=F honorCodec=F EOType=0 DTMF Caps(1,2,101,1,T) confID=0 connType=3 connStatus=0 mtpPre=F teleEve=0 IFCreated=F IFHandling=0
FS=0 mcNodeId=0LatentCaps=null dtm.mode=0 dtm.CI=0 dtm.MTPForDTMF=F reConnType=0 videoCall=F AllowedCallType=0x0 mtpChanged=F preCLvl=5 resCap=0
party1.mMediaCoordinatorNodeId=0 party2.mMediaCoordinatorNodeId=0 sideBAns= T
```

```
00029255.001 |19:38:55.395 |AppInfo |SIG-MediaManager-(6)::wait_MediaConnectRequest, CI(22104488,22104489), capCount(7,1), mcNodeId(0,0), xferMode(16,16), reConnectType(0), mrid (0,
22104490) IFCreated(0 0) proIns(0 0), AC(0,0), party1DTMF(1 3 101 0 0)party2DTMF(1 2 101 1 1),reConnFlag=0, connType(3,3), IFHand(0,0),MTP(0,2),MRGL(893975db-2dc3-040e-1fb0-
36c7ffe0598b,e3e55643-ff44-a1fe-badc-539e66a2e423) videoCap(0 0), mmCallType(0),FS(0,0), IpAddrMode(0 0) aPid(1, 82, 11), bPid(1, 82, 12) EOType(0 0) () honorCodec(0 0)
```

- Party1, Party2: Media Connect Device, Trunk pair
- CI
- region: set Region

- xferMode: Interface type such as Device or Trunk

xferMode: interface type such as Device or Trunk

Device / Interface	xferMode
SCCP Endpoint	4
MOH	4: Unicast MOH, 9: Multicast MOH
H323 Endpoint (Gateway, POTS)	7
MTP/Transcoder	10
MGCP	12
H323 Trunk	13
SIP Endpoint/Trunk	16

- DTMF Caps(dtmfConfig, supportedDTMFMethod, rfc2833RecvPayloadNum, wantDTMFReception, provideOOB)
- dtmfconfig: CUCM Admin GUI SIP Trunk configuration

dtmfConfig	Description
BestEffort = 1	A setting that uses the matched DTMF method. It is an initial value and is fixed at BestEffort except for SIP Trunk.
PreferOOB = 2	Configuration to use OOB if SIP Trunk supports it.
Prefer2833 = 3	Configuration to use 2833 if SIP Trunk supports it.
PreferBoth = 4	A setting that uses both OOB and 2833 if SIP Trunk supports it.

- provideOOB: T = provide OOB

- Supported DTMF Method:
- NODTMF=0
- OOB=1
- RFC2833=2
- OOB and RFC2833=3
- Unknow DTMF=4

- Party1: DTMF Caps (1 3 101 0 F), Party2: DTMF Caps (1 2 101 1 T) -> Party 1 is OOB and 2833, Party 2 is 2833 (Media Termination Point Required in SIP Trunk settings) If it is included and the DTMF Signaling Method is No Preferences, SIP Trunk will only negotiate with RFC2833) but MTP is required as it also needs to provide OOB.

- MTP (0,2) : Party1 is 0 (no MTP required), Party2 is 2 (RequiresRFC2833MediaTerminationPoint)

MTP Requirements	Description
NoRequirements = 0	MTP don't
RequiresMediaTerminationPoint = 1	MTP necessary
RequiresRFC2833MediaTerminationPoint = 2	MTP required for 2833
RequiresMediaTerminationPointAndUseTRP = 3	Both "MTP required" and "Use TRP" are checked
RequiresRFC2833MediaTerminationPointAndUseTRP = 4	Both "MTP required" with RFC2833 cap and "Use TRP" are checked
RequiresRFC2833AndUseTRP = 5	"Use TRP" is checked, and RFC2833 cap needed
UseTRP = 6	"Use TRP" is checked
Disable_AudioPassThru = 7	Media will disable audio pass thru if either endpoint sets this flag.

- If there is a mismatch between the 2833 and the OOB, allocate the MTP on the side that supports the 2833. In this state, if another factor requires her MTP, reuse his assigned MTP.